

EX NO: 1b: CHARACTERISTICS STUDY OF SIDE BAND POWER AND FREQUENCIES IN AM USING MATLAB

Objectives:

- i) To generate an Amplitude Modulated (AM) signal in MATLAB using specified carrier and modulating signal parameters.
- ii) To determine and verify the upper sideband (USB) and lower sideband (LSB) frequencies of the AM signal using frequency spectrum analysis in MATLAB.
- iii) To calculate the carrier power, sideband power, and total transmitted power of the AM signal for a given modulation index using MATLAB.
- iv) To analyze the effect of varying the modulation index on sideband power distribution and overall transmission efficiency through MATLAB simulation.

PROGRAM (MATLAB CODE):

```
clc;  
  
clear;  
  
close all;  
  
Am=1;  
  
fm=500;  
  
Ac=5;  
  
fc=5000;  
  
fs=20*fc;  
  
t=0:1/fs:4/fm;
```

```

m = Am*sin(2*pi*fm*t);

c = Ac*sin(2*pi*fc*t);

mu = Am/Ac;

am = Ac*(1+mu*sin(2*pi*fm*t)).*sin(2*pi*fc*t);

dem = abs(hilbert(am)); dem = dem-mean(dem);

N=length(am); f=linspace(-fs/2,fs/2,N);

AM_FFT=abs(fftshift(fft(am)))/N;

USB=fc+fm; LSB=fc-fm;

Pc=Ac^2/2; Psb=(mu^2*Pc)/4;

figure('Name','AM Modulation','NumberTitle','off');

subplot(3,2,1), plot(t,m), title('Message'), grid on

subplot(3,2,2), plot(t,c), title('Carrier'), grid on

subplot(3,2,3), plot(t,am), title('AM Signal'), grid on

subplot(3,2,4), plot(t,dem), title('Demodulated'), grid on

subplot(3,2,5), plot(f,AM_FFT), xlim([fc-3*fm fc+3*fm]), title('Sidebands'), grid on

subplot(3,2,6), bar([Pc Psb Psb]), set(gca,'XTickLabel',{'Carrier','USB','LSB'}),

title('Power'), grid on

fprintf('μ=%.2f | USB=%d Hz | LSB=%d Hz | Pc=%.2f W | Psb=%.3f

W\n',mu,USB,LSB,Pc,Psb);

```

Output:

